

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.4: Transport in Plants — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.4: Transport in Plants
Driving Question	DRIVING QUESTION: How does a maize plant in a Kenyan farm move water and nutrients from the soil all the way to its topmost leaves — and what happens to that water in the end?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon: The Travelling Red Dye & The Sweating Bag	A maize seedling from the Kisumu school garden with its roots in red-dyed water: after 30 minutes, a cross-sectional cut revealed vivid red rings only in specific dot-shaped woody regions of the stem — the outer green tissue and central pith remained completely white. Simultaneously, water droplets formed on the inside of a sealed polythene bag around a sweet potato cutting placed in afternoon sun, even though no liquid water had been added to the bag.	Water does not move randomly through the entire stem — it travels through specific, distinct channels. The leaves of a plant release water vapour even when the plant appears dry on the outside. These two observations together suggest that plants actively move water in a directed pathway from roots to leaves, and that water exits the plant through the leaves in a form we cannot see until it condenses.	No full explanation yet — this lesson opens the inquiry. Learners have generated initial models and posted their questions on the Driving Question Board. The key questions to be resolved across the sub-strand are: What are those ring-shaped channels and how do they pull water upward against gravity? Why does the plant release water through its leaves even when it needs water to survive? These questions will be answered through the laboratory and observation activities in Lessons 2–5.
Lesson 2 Structures for Absorption: Root Hair Cells, Xylem & Phloem, and Capillary Forces	Photomicrographs of root hair cells and transverse sections of root and stem showing xylem and phloem; a live capillary-tube demonstration using red food dye showing water rising higher in narrower tubes; six gallery-walk diagrams showing cell structures and molecular forces.	Root hair cells have a large surface area that maximises water absorption from the soil. Xylem vessels are narrow, hollow, dead cells arranged in rings or bundles in the stem — these are the channels the red dye travels through in the phenomenon. Phloem is distinct from xylem and carries dissolved sugars. Adhesion (water sticking to xylem walls) and cohesion (water molecules sticking to each other) are the physical forces that help water rise in narrow tubes — just as water rises higher in a narrower capillary tube.	The red-stained rings in the cross-sectionally cut maize stem from Kisumu are xylem vessels — not phloem or any other tissue. Water is able to enter these narrow vessels partly because of adhesive and cohesive forces, analogous to the capillary rise seen in the demonstration. Root hair cells increase the surface area available for water absorption from the soil. These structural observations begin to explain how water moves upward against gravity, though the full mechanism (cohesion-tension theory, transpiration pull) will be

			established in Lesson 7.
Lesson 3 Absorption of Water and Mineral Salts: Osmosis, Active Transport, and Root Pressure	In Lesson 1 we observed that red food dye placed in a beaker of water entered the roots of a freshly uprooted maize seedling from our Kisumu school garden and moved upward, appearing as vivid red rings only inside the xylem tissue when the stem was cut cross-sectionally. In Lesson 2 we identified the xylem and phloem as the transport channels. In today's lesson we examined diagrams and research resources showing root hair cells in contact with soil water, and we observed that the cell sap inside the root hair cell vacuole is more concentrated (lower water potential) than the water in the surrounding soil.	Water enters root hair cells by OSMOSIS — it moves from a region of higher water potential (dilute soil water) to a region of lower water potential (concentrated cell sap) across the semi-permeable cell membrane. Mineral salts such as phosphate ions from DAP fertiliser and ammonium ions from CAN or Urea cannot enter by osmosis; they are absorbed by ACTIVE TRANSPORT — protein carrier molecules in the cell membrane use ATP energy to pump ions against their concentration gradient into the root hair cell. As water and mineral ions accumulate in the root cells and pass inward through the cortex to the xylem, they create a hydrostatic pressure called ROOT PRESSURE, which pushes water upward into and through the xylem vessels. This is why the red dye in our phenomenon moved upward against gravity. Over-application of fertiliser increases solute concentration in the soil, reversing the water potential gradient and causing water to leave the root cells (plasmolysis), which is why sukuma wiki or maize can wilt after excess fertiliser is applied.	The red dye entered the maize root and rose through the xylem because: (1) osmosis drove water molecules from the relatively dilute soil water into the more concentrated root hair cell vacuole across the semi-permeable membrane; (2) active transport used ATP to pump mineral ions such as phosphate (from DAP) into the root cells against their concentration gradient, further lowering water potential and sustaining the osmotic gradient; and (3) the accumulation of water in the xylem generated root pressure — a hydrostatic push that propelled the dye-coloured water upward through the xylem channels, which is why only the xylem rings (not the phloem or pith) turned red. Our model now explains the ROOT END of the journey. However, root pressure alone may not be sufficient to lift water to the topmost leaves of a two-metre maize plant — this is the unresolved question we carry into Lesson 7.
Lesson 4 Translocation: How Does the Maize Plant Move Sugar from Leaf to Root?	We observed a ring-barked (girdled) bean or sweet-potato stem where a band of outer tissue (phloem) was removed: tissue swelled and accumulated just ABOVE the cut but not below. We also analysed Evidence Cards showing sieve tubes, companion cells, and aphid stylet experiments confirming sugary sap flows under pressure through phloem.	Plants have TWO transport systems. Xylem carries water and mineral salts upward from roots to leaves (as shown by the red-dye experiment in Lessons 1–3). Phloem carries sugars and other assimilates by TRANSLOCATION from SOURCE regions (mature photosynthesising leaves) to SINK regions (roots, developing maize cob, young growing leaves) — the direction is always from where sugar is made or stored toward where it is needed. Ring-barking blocks phloem but not xylem, explaining why girdled trees eventually die even though water still reaches their leaves.	The red-stained rings in our Kisumu maize stem cross-section (the anchoring phenomenon) are XYLEM — they carry only water and mineral salts upward. The colourless outer tissue of the vascular bundle is PHLOEM — it carries sugars made by photosynthesis to the rest of the plant by translocation. Together, these two systems explain how the maize plant sustains every living cell: xylem supplies water and minerals for photosynthesis; phloem distributes the glucose produced. The polythene-bag water droplets (transpiration, second part of the phenomenon) will be fully explained in Lesson 5 — that water exits through stomata from the xylem stream.

Lesson 5 Transpiration Experiment Setup: Designing Investigations into Factors Affecting Water Loss in Plants	A freshly cut sweet-potato or bean stem was placed in a beaker of water. Different groups changed one factor each: Group A coated leaf undersurfaces with petroleum jelly; Group B directed a fan at the shoot; Group C enclosed the shoot in a sealed clear polythene bag; Group D placed their setup in full Kisumu afternoon sunlight versus shade. Each group recorded the initial water volume at Time = 0.	Transpiration — the loss of water vapour from plant surfaces — can be affected by leaf surface conditions (open or blocked stomata), wind speed, surrounding humidity, and light intensity. A fair test requires changing only ONE variable at a time while keeping all others constant. The rate of transpiration is measured by how much water the plant loses over time.	Stomata are small pores, mostly on the underside of leaves, through which water vapour diffuses out of the plant into the surrounding air. The rate of this diffusion depends on the concentration gradient of water vapour between the inside of the leaf and the air outside. Anything that increases that gradient (wind removing vapour, bright light opening stomata wider) speeds transpiration; anything that reduces it (high humidity in a sealed bag, petroleum jelly blocking pores) slows it. This is the final stage of the water journey that began when roots absorbed water from the Kisumu farm soil and xylem carried it to the topmost leaves.
Lesson 6 Transpiration Data Collection and Analysis: What Patterns Does Our Evidence Reveal?	We observed that the group with high wind lost the most water, the group with blocked stomata (petroleum jelly) lost the least, the group in the sealed polythene bag lost very little, and the group under bright light lost more than the group in shade. We pooled all four group results into one class data table and drew bar graphs showing these differences.	We learned that the rate of transpiration is controlled by the concentration gradient of water vapour between the inside of a leaf and the surrounding air. Wind increases the rate by sweeping away water vapour outside the stomata, restoring a steep gradient. High humidity reduces the rate by raising the water vapour concentration outside, flattening the gradient. Bright light increases the rate by causing stomata to open wider. Blocking stomata reduces the rate almost to zero.	We explained why water droplets formed inside the sealed polythene bag in Lesson 1: the leaf was continuously losing water vapour through its stomata by diffusion down the concentration gradient, and when the bag trapped that vapour it condensed on the cool inner surface. This shows that transpiration is a continuous, measurable process driven by physical gradients — not a random event — and its rate changes predictably when environmental conditions change.
Lesson 7 Importance of Transpiration and Community Application	In earlier lessons, learners placed a maize seedling's roots in red-dyed water and observed vivid red rings appearing only in the xylem tissue of a cross-sectioned stem. They also observed water droplets forming on the inside of a sealed polythene bag enclosing a leafy plant cutting placed in afternoon sun — even though no liquid water was added to the bag.	Transpiration serves three critical functions for the maize plant: (1) it cools the leaf surface through evaporation, preventing heat damage on hot Kenyan afternoons; (2) it generates the transpiration pull (cohesion-tension mechanism) that is the primary force driving water upward through the xylem against gravity — explaining why the red dye moved upward through those specific channels; and (3) it drives the continuous transpiration stream that carries dissolved mineral salts (such as nitrogen from CAN/Urea and phosphorus from DAP) from the soil, through the roots, and all the way to the topmost leaves. At the landscape scale, transpiration from forests like the Mau Forest feeds water vapour back into the atmosphere, sustaining rainfall	The water that entered the maize plant through its root hair cells — visible as the red dye moving through the xylem rings — ultimately escapes as water vapour through the stomata of the topmost leaves (the water droplets seen on the inside of the polythene bag). This loss is not waste: it is the engine of the entire transport system. Without transpiration pulling water upward through the cohesion-tension mechanism, mineral nutrients dissolved in the soil water could not be delivered to the growing tissues. Responsible farming practices — regular watering to maintain the water column, manuring to improve soil structure and water retention, and careful use of fertilisers (DAP for phosphorus and nitrogen, CAN and Urea for nitrogen) —

		patterns that support farming communities across western Kenya and the Lake Victoria basin.	directly support this transport system and sustain maize crop yields on Kenyan farms.
Lesson 8 MODEL BUILDING AND DQB CLOSURE: The Full Journey of Water in a Maize Plant	We observed red dye moving upward through the xylem rings of a cut maize stem cross-section, water droplets forming on the inside of a polythene bag enclosing a leafy cutting, transpiration rates changing with wind speed, humidity, and light intensity in our group experiments, and wilting and recovery in plants under water stress — all real phenomena visible in Kenyan school gardens and farms.	We learned that water enters root hair cells by osmosis down a water potential gradient, that mineral salts enter by active transport requiring energy, that water travels up the xylem — not the phloem — driven by cohesion-tension and transpiration pull created at the leaf surface, that sugars travel downward and upward through the phloem by translocation from source to sink, that transpiration cools leaves, sustains the transpiration stream carrying minerals to the topmost leaves, and contributes water vapour to regional water cycles such as those sustained by the Mau Forest.	We explained why only the xylem rings turned red in the maize stem cross-section, why water droplets formed on the polythene bag from evaporation through stomata, why a wilting plant in Mombasa recovers after evening watering through osmosis restoring cell turgor, why DAP fertiliser applied to a Kisumu maize farm reaches the topmost leaves via the transpiration stream, and why cutting a ring of bark around a stem (girdling) kills the plant above the cut by blocking phloem translocation while leaving xylem intact.

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